

This policy is currently under review and the current version remains endorsed until the review is complete – September 2022

Guideline

Transient Ischaemic Attack (TIA) and Stroke Guideline

Site	Department	Disciplines
Armada Health Service	Emergency Department	Medical

Acute Stroke

Possible acute stroke/TIA in patients with symptoms < 4 hours:

- Hemiparesis
- Hemisensory symptoms
- Dysphasia
- Ataxia
- Diplopia and Dysarthria
- Sudden visual loss

***Immediately notify senior ED doctor**

***Urgent CT Head**

***Discuss with RPH neurology if potential for thrombolysis**

If stroke symptoms present but > 4 hrs post onset

- exclude haemorrhage
- seek and treat immediate life threats
- exclude stroke mimics eg migraine, hypo/hyperglycaemia, seizure, sepsis, tumour, syncope, peripheral vestibulopathy (see [Appendix 1: ROSIER tool](#))

Evaluation

- Bedside: BSL, ECG, U/A
- Laboratory: FBC, UEC, LFTs, PT/INR (if on warfarin)
- CT brain (non-contrast), CXR

Treatment

- Aspirin 100mg if no haemorrhage on CT
- NBM until bedside swallow assessment
- Instigate ongoing physiological monitoring and treatment:
 - Neurological status
 - Blood pressure
 - General consensus is for markedly elevated blood pressure (BP >220/120) to be cautiously reduced (by no more than 10-20%) and patient observed for neurological deterioration done in conjunction with Senior Dr and RPH Neurology Team
 - Avoid hypotension
 - Blood glucose
 - Avoid hypoglycaemia

- Management of markedly elevated glucose should be done in conjunction with RPH Neurology Team
- Hydration status

Transient Ischemic Attacks (TIAs)

- Transient ischemic attack (TIA) is defined as a transient episode of neurologic dysfunction caused by focal brain, spinal cord, or retinal ischemia, without acute infarction
- The risk of stroke after a TIA is 3.5%, 8% and 9.2% at 2, 30 and 90 days respectively

Approximately **half of the early risk of stroke after TIA is seen within the first 48 hours**, early diagnostic workup and early treatment is necessary to prevent further events

- Patient with TIA should have full assessment which includes Stroke risk stratification using ABCD² Tool score
- Patients should be classified as high or low risk, and further investigations and management done based on their classification
- The two-day risk of stroke is 1% for an ABCD² score of 0 - 3 but 4.1% for a score of 4 - 5 and 8.11% for a score of 6 – 7
- 20% of patients with ABCD² <4 had >50% carotid stenosis or atrial fibrillation

Guideline

ABCD ² Tool			Points
A	Age	≥ 60 years	1
B	Blood Pressure	SBP> 140MMhG, or DBP> 90MMhG	1
C	Clinical Features	Any unilateral weakness	2
		Speech impairment without weakness	1
D	Duration	≥ 60 minutes	2
		10 -59 minutes	1
D	Diabetes	Diabetic	1

ABCD ² Tool Interpretation	Points
High Risk*	≥ 4
Low Risk	≤3

***Patients are also considered High Risk if there are other high risk features:**

- New or paroxysmal atrial fibrillation (AF).
- Tight carotid stenosis (audible carotid bruit).
- Two or more TIAs in previous week (crescendo TIA).
- Known or suspected hypercoagulable state.
 - Age 45 or younger.
 - History of clotting dysfunction.

- Multiple venous or arterial occlusions.
- Suspected or confirmed cancer.
- Family history of hypercoagulable state.

Investigations

All TIA Patients	• FBC, UEC, LFTs, BSL, PT / INR • ECG • Urgent CT Head
High Risk Patients	• Urgent carotid duplex ultrasound (patients with carotid territory symptoms)* • Urgent Echocardiogram #
Low Risk Patients	• Carotid duplex US / Echocardiogram # As soon as possible (within 48 hours)

*E.g. amourosis fugax, dysphasia.

Patients with AF/ heart failure / recent (<3 months) AMI / cardiac murmur / peripheral signs of embolism / clinical events in >2 territories (right, left hemisphere or anterior, posterior circulation).

Treatment

Antithrombotic treatment if patient has no bleed or mass on CT Head	Aspirin 100 mg stat, then 75- 100 mg OD* *Preferred in acute TIA / stroke
Patient on daily aspirin	Asasantin SR 200/25 mg BD or Stop aspirin and start clopidogrel 75mg / day
Aspirin allergy or cannot tolerate	Clopidogrel 75mg / day
Not recommended	Aspirin and clopidogrel combination (unless the patient has concomitant coronary artery disease or a recent coronary stent)

Disposition

High Risk Patients	• Admit to the Medical Ward or Stroke Unit • Admit team to arrange carotid Duplex, echocardiogram, and MRI as warranted
Low Risk Patients	• Discharge home post CT Head ± Carotid Duplex US • Start patient on antithrombotic treatment • Follow up with their GP ASAP to schedule outpatient investigations (echocardiogram + /- Carotid Duplex)

Related Standards

NSQHS

Standard 1: Governance for Safety and Quality in Health Service Organisation

Standard 4: Medication Safety

EQulP National

Standard 12: Provision of Care

References

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4. Stroke Foundation. (2017). *Clinical guidelines for stroke management 2017*. Melbourne, Australia. Retrieved from <https://informme.org.au/Guidelines/Clinical%20Guidelines%20for%20Stroke%20Management%202017>
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Compiled by:

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Authorisation

Authorising Committee	Date Authorised	Date of Next Review
Director Emergency Services AHS	2011, August 2014	August 2017

Stroke and Transient Ischaemic Attack (TIA) Guideline

Emergency Department Safety and Quality Meeting	13/12/2017	13/12/2020
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Appendix 1: Recognition of Stroke in the Emergency Room (ROSIER)

*This is a tool for use in the ED for recognition of stroke and helps identify mimics

GCS E=___ M=___ V=___ BP= ___ / ___ *BG= _____

*If BG < 3.5 mmol/L, treat urgently and reassess once blood glucose normal

Has there been loss of consciousness or syncope? Y (-1) ☐ N (0) ☐

Has there been seizure activity? Y (-1) ☐ N (0) ☐

Is there a NEW ACUTE onset (or on awakening from sleep)

I. Asymmetric facial weakness Y (+1) ☐ N (0) ☐

II. Asymmetric arm weakness Y (+1) ☐ N (0) ☐

III. Asymmetric leg weakness Y (+1) ☐ N (0) ☐

IV. Speech disturbance Y (+1) ☐ N (0) ☐

V. Visual field defect Y (+1) ☐ N (0) ☐

Total Score _____ (-2 to +5)

Provisional diagnosis

☐ Stroke ☐ Non-stroke (specify) _____

Note: Stroke is unlikely, but not completely excluded if total scores are ≤ 0 .

**Appendix 2: Management of a Transient Ischaemic Attack (TIA) in ED
Flowchart**

